BCSE355L: Cloud Architecture Design

Final Project Comprehensive AWS Architecture & Engineering Defense Report

Project: CloudShield (Predictive Disaster Logistics)	**AWS Account ID:** 206238482111
AWS Region: ap-southeast-2 (Sydney)	**Target Host:** http://32.236.173.240:8000
Syllabus Version: 2.0 (8 Specialized Modules)	**Live Mode:** AWS_MODE=live (IMDSv2 Enforced)
GitHub Repository: https://github.com/NeeleshUnleashed/CloudShield	**Status:** 100% Deployed & E2E Verified (12/12 Tests)

1. Executive Summary & Curriculum Alignment

This project was engineered to directly satisfy the requirements of **BCSE355L (Cloud Architecture Design)**. CloudShield is an autonomous, explainable disaster response resource allocation platform combining deterministic multi-criteria risk scoring (DPRI), Sphere Humanitarian Standards demand forecasting, and multi-depot optimization with a two-stage human-in-the-loop approval gate. The application is fully operational in the Sydney AWS region (ap-southeast-2) adhering strictly to the AWS Well-Architected Framework.

2. Comprehensive AWS Services Breakdown (What, How & Why)

Each AWS service deployed in CloudShield directly corresponds to a specific unit of the BCSE355L syllabus:

Syllabus Module	AWS Service Used	Implementation (HOW it is used)	Architectural Rationale (WHY needed)
Module 1 Cloud Foundations & IAM	**AWS IAM** Role & Profile	CloudShieldEC2AppRole attached via CloudShieldEC2Profile. Enforces least-privilege inline policy for SNS, S3, and CloudWatch. Uses IMDSv2 for ephemeral credential rotation.	Eliminates static hardcoded API keys. Prevents credential leakage in GitHub/disk. Limits blast radius if the instance is compromised.
Module 2 Compute & Storage	**Amazon EC2** (Compute)	t3.micro (2 vCPUs, 1GB RAM) running Amazon Linux 2023. Managed as a systemd background daemon (cloudshield.service) running FastAPI on port 8000.	Eliminates cold-start latency for real-time disaster triage. In ap-southeast-2, t3.micro is Free Tier eligible (750 hrs/mo free) while t2.micro is not.
Module 2 Compute & Storage	**Amazon EBS** (Block Store)	8 GB gp3 General Purpose SSD root volume (/dev/xvda). Baseline 3,000 IOPS, 125 MB/s throughput. Hosts OS and active SQLite transactional database.	Provides persistent, low-latency POSIX file locks necessary for ACID database transactions. Configured with DeleteOnTermination: true to avoid orphaned charges.
Module 2 Compute & Storage	**Amazon S3** (Object Store)	cloudshield-artifacts-206238482111-ap-southeast-2. Stores automated atomic SQLite database snapshots. Enforces 7-day expiration lifecycle & Block Public Access.	Guarantees 11 9's durability (99.999999999%) for disaster records. The 7-day lifecycle prevents backup accumulation and guarantees zero cost overages.
Module 3 Networking & Content Delivery	**Amazon VPC & Security Groups**	VPC vpc-02a4493bff64cb2c9, public subnet in ap-southeast-2a. Security group cloudshield-app-sg restricts TCP port 8000 exclusively to 182.66.218.113/32.	Hardware-level stateful perimeter filtering. Protects lightweight demo RBAC from public internet scrapers and bots. Avoided $32.85/mo NAT Gateway charge trap.
Module 4 Database Services	**Database Architecture** (EBS/RDS/DynamoDB)	Deployed with ACID SQLite on EBS + S3 sync. Documented migration path to **Amazon Aurora PostgreSQL Multi-AZ** and **Amazon DynamoDB**.	SQLite eliminates $15+/mo RDS idle instance charges for college demo while guaranteeing full ACID serializability. Enterprise blueprint details Multi-AZ failover.
Module 5 Management & Cost Optimization	**Amazon CloudWatch Logs**	Log Group /cloudshield/application with streams app-stdout and app-stderr shipped by CloudWatch Agent. Enforces strict 1-day retention.	Centralized forensic log streaming outside the host. 1-day retention guarantees log volume remains well beneath the 5 GB Free Tier monthly ceiling.

Module 6 Security & Compliance	**Security & IMDSv2**	Enforced HttpTokens=required and HttpPutResponseHopLimit=2 on EC2. Implements the AWS Shared Responsibility Model.	Neutralizes SSRF (Server-Side Request Forgery) attacks by requiring a session token header, preventing credential theft from internal metadata services.
Module 7 Application Integration	**Amazon SNS** (Pub/Sub)	Standard topic CloudShield-Disaster-Alerts. Invoked via boto3 client upon LOGISTICS_DIRECTOR disaster plan approval.	Decoupled publish/subscribe architecture. Fans out critical disaster supply dispatches to field teams, hospitals, and email/SMS subscribers concurrently.

3. How AWS Works in the Backend — End-to-End Execution Trace

When an incident is processed in CloudShield, data flows across 8 discrete cloud subsystems:

Step 1: Network Ingress & Hardware Inspection

The user connects to http://32.236.173.240:8000. The AWS edge router routes packets to vpc-02a4493bff64cb2c9. Security Group cloudshield-app-sg inspects the source IPv4. Packets from 182.66.218.113/32 pass; all others are dropped at the hypervisor level.

Step 2: EC2 Host & Systemd Daemon Supervision

The Linux kernel delivers the stream to port 8000. Supervised by cloudshield.service, Uvicorn processes the request. If the application process encounters an unhandled exception, systemd restarts it in 5 seconds automatically.

Step 3: IMDSv2 Authentication Handshake

When Python initializes boto3.client('sns'), it issues a PUT request to 169.254.169.254/latest/api/token to fetch a cryptographic session token, and retrieves rotating temporary STS credentials for CloudShieldEC2AppRole. No static keys exist on disk.

Step 4: Deterministic DPRI & Humanitarian Demand Prediction

The core engine calculates the Disaster Priority Risk Index: DPRI = [(Hazard * Multiplier) + (Vuln * 0.3) + ((1 - Coping) * 0.25) + (Infra * 0.2)] / 2.05. It estimates supply requirements for Water (15L/day), Food (1 pack/day), Medical, Shelter, and Power according to Sphere Standards.

Step 5: Two-Stage Human-in-the-Loop Approval Gate

The incident is persisted with status PENDING_APPROVAL. **Zero depot inventory is deducted**. Only an authorized user with role LOGISTICS_DIRECTOR can approve the plan. Unauthorized roles receive **HTTP 403 Forbidden**.

Step 6: Atomic Database Transaction on EBS Volume

Upon Director approval, an atomic SQLite transaction decrements regional stock (Sydney, Brisbane, Melbourne, Perth), records the dispatch manifest, and sets status to APPROVED_DISPATCHED. Duplicate approvals are rejected with **HTTP 400 Bad Request**.

Step 7: Real-Time Amazon SNS Event Broadcasting

The approval service publishes an event payload to arn:aws:sns:ap-southeast-2:206238482111:CloudShield-Disaster-Alerts. SNS broadcasts the message to subscribers, returning a unique MessageId stored in the audit history.

Step 8: CloudWatch Log Streaming & S3 Snapshot Backup

The CloudWatch Agent tails /var/log/cloudshield.log and streams events to /cloudshield/application. Concurrently, an hourly cron job executes sqlite3 .backup and uploads database snapshots to S3.

4. AWS Well-Architected Framework (6 Pillars Defense)

Module 7 requires demonstration of the six pillars of the AWS Well-Architected Framework:

Pillar	CloudShield Implementation & Design Choice
1. Operational Excellence	Infrastructure-as-Code via automated bash bootstrap; CloudWatch centralized logging; systemd automatic process recovery; S3 automated backups; /api/health probes.
2. Security	IMDSv2 session tokens; least-privilege IAM instance role; /32 restricted Security Group; S3 Block Public Access; zero secrets in source code; backend RBAC.
3. Reliability	Atomic ACID transactions on EBS; stateless API tier; S3 99.999999999% durability; automatic systemd restart within 5s; deterministic mathematical models.
4. Performance Efficiency	Python 3.11 asynchronous event loop; zero network database latency (local EBS); burstable T3 CPU architecture; low-latency in-memory prediction calculations.

5. Cost Optimization	Avoided $32.85/mo NAT Gateway; avoided $16.50/mo ALB; avoided $15/mo RDS; t3.micro Free Tier eligible in Sydney; 1-day CloudWatch retention; 7-day S3 lifecycle.
6. Sustainability	Right-sized micro instance; minimal package dependencies; idempotent single-command decommissioning scripts to eliminate idle cloud energy waste.

5. College Viva Defense — High-Weightage Examination Q&A;

Expected questions from the BCSE355L examination committee with exact model answers:

Q1: Why didn't you deploy an Application Load Balancer (ALB) or NAT Gateway?

Model Answer: In AWS, an ALB costs ~$16.50/month and a NAT Gateway costs ~$32.85/month plus data processing fees. In BCSE355L Module 5 & 7, Cost Optimization is a core pillar. For a single-instance disaster intake node, routing traffic through a direct Internet Gateway with a stateful Security Group locked to our /32 CIDR provides identical security isolation while eliminating over $50/month in idle cloud expenditure.

Q2: How does IMDSv2 protect your EC2 instance compared to IMDSv1?

Model Answer: IMDSv1 relied on unauthenticated HTTP GET requests to 169.254.169.254, making it vulnerable to Server-Side Request Forgery (SSRF) attacks. CloudShield explicitly configured HttpTokens=required (IMDSv2). This enforces session-oriented PUT requests with custom header tokens, preventing SSRF exploits and unauthorized extraction of temporary IAM instance credentials.

Q3: What is the architectural difference between your Security Group and a Network ACL?

Model Answer: A Security Group is stateful and operates at the virtual network interface (ENI) level; allowing inbound traffic on port 8000 automatically permits the return response. A Network ACL (NACL) is stateless and operates at the subnet boundary, requiring explicit rules for both inbound traffic and ephemeral outbound ports (1024-65535). CloudShield utilizes Security Groups for fine-grained host protection.

Q4: Why did you choose t3.micro instead of t2.micro for the Sydney region?

Model Answer: Querying the AWS CLI in ap-southeast-2 (aws ec2 describe-instance-types) reveals that t2.micro is NOT Free Tier eligible in Sydney, whereas t3.micro is explicitly Free Tier eligible (750 hours/month). Furthermore, T3 runs on the modern AWS Nitro System with 2 vCPUs and Enhanced Networking (ENA), providing significantly better burstable throughput than T2.

Q5: How does your S3 Lifecycle Policy prevent cloud billing overrun?

Model Answer: Our EC2 instance captures hourly backups of the SQLite database. Over several months, unmanaged backups would exceed the 5 GB Free Tier limit. We deployed an S3 lifecycle configuration that permanently deletes backup objects after 7 days, maintaining steady-state storage under 100 MB.

Q6: How does CloudShield comply with the AWS Shared Responsibility Model?

Model Answer: Under the Shared Responsibility Model (Module 1 & 6), AWS manages Security OF the Cloud (physical data centers in Sydney, Nitro chips, power, and hypervisors). We as the customer manage Security IN the Cloud: configuring least-privilege IAM policies, restricting Security Group ingress, patching the Amazon Linux OS, enforcing IMDSv2, and managing application-level RBAC authorization.

6. Verified Post-Demo Cleanup Protocol

To decommission all AWS resources and return the account to $0.00 recurring charges after your presentation:

```
# 1. Terminate EC2 Instance & wait
aws ec2 terminate-instances --instance-ids i-0f5a671d1b8a31d23 --region ap-southeast-2
aws ec2 wait instance-terminated --instance-ids i-0f5a671d1b8a31d23 --region ap-southeast-2

# 2. Delete Security Group
aws ec2 delete-security-group --group-id sg-08249050614cb6db9 --region ap-southeast-2

# 3. Empty & Delete S3 Bucket
aws s3 rm s3://cloudshield-artifacts-206238482111-ap-southeast-2 --recursive --region ap-southeast-2
aws s3api delete-bucket --bucket cloudshield-artifacts-206238482111-ap-southeast-2 --region ap-southeast-2

# 4. Delete SNS Topic & CloudWatch Log Group
aws sns delete-topic --topic-arn arn:aws:sns:ap-southeast-2:206238482111:CloudShield-Disaster-Alerts --regio
n ap-southeast-2
aws logs delete-log-group --log-group-name /cloudshield/application --region ap-southeast-2

# 5. Delete IAM Instance Profile & Role
aws iam remove-role-from-instance-profile --instance-profile-name CloudShieldEC2Profile --role-name CloudShi
eldEC2AppRole
aws iam delete-instance-profile --instance-profile-name CloudShieldEC2Profile
aws iam delete-role-policy --role-name CloudShieldEC2AppRole --policy-name CloudShieldAppPolicy
aws iam delete-role --role-name CloudShieldEC2AppRole
```